Lecture 1: Course introduction

PSYC 51.07: Models of language and communication

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About the instructor

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Research interests:

- Memory and learning
- Brain network dynamics
- Natural language processing

🧠 Memory

📊 Data science

🤖 NLP

> **Note**
>
> **Lab website:** context-lab.com | **GitHub:** ContextLab

What is this course about?

Note

We will explore how machines can understand and generate human language:

- Building conversational agents from scratch
- Understanding how language relates to thought
- Hands-on programming with real models
- Critical thinking about AI consciousness

Warning

This course is experiential! You will learn by doing: coding, experimenting, discussing, and researching.

Course structure

ELIZA → SPAM → Wikipedia → Chatbot → GPT

Five projects, building from pattern matching to transformer models

Note

Grading

- Problem sets (5): 75%
- Final project: 25%

Tip

Tools

- Google Colaboratory
- HuggingFace
- GitHub / Discord

The big questions

Throughout this course, we will grapple with:

1. Can machines truly *understand* language?
2. What is the relationship between language and thought?
3. How do statistical patterns capture meaning?
4. Can AI be conscious?

These are not just philosophical questions--they are at the heart of cognitive science!

Discussion: Is ChatGPT conscious?

> **Note**
>
> **Discussion:** Let's start with a provocative question: **Is ChatGPT conscious?**

Think about:

- What does "conscious" even mean?
- How would we test for consciousness?
- Does it matter if ChatGPT *seems* conscious?

What is consciousness?

Perspectives:

Phenomenal | Access | Full

- **Phenomenal:** Subjective experience
- **Access:** Information for reasoning
- **Self-awareness:** Knowledge of mental states

If ChatGPT says "I feel happy," does it actually *feel* anything?

The hard problem

With humans, we assume consciousness because:

- We share similar biology
- We have our own conscious experience
- They behave consistently with having experiences

But with AI:

- Completely different "biology" (silicon vs. neurons)
- No shared evolutionary history
- Can produce human-like behavior

> **Note**
>
> **Why is this so difficult?** We cannot directly observe consciousness--even in other humans!

The Chinese room argument

> **Note**
>
> **John Searle (1980):** Thought experiment about understanding vs. simulation

The setup: Person in room with Chinese symbols and rule book

Chinese question → Rule book → Chinese answer

Person follows rules but does not understand Chinese

> **Tip**
>
> Does following rules = understanding? Searle

Current scientific consensus

> Warning
>
> **Important:** Most cognitive scientists and AI researchers agree: **Current LLMs are not conscious.**

Why not?

- No sensory-motor grounding in the world
- No persistent self-model or goals
- Pattern matching $\neq$ understanding

Language and thought

> **Note**
>
> **Discussion:** Do you need language to think? Does language *shape* how you think?

Language = Thought:
All thinking happens in language

Language ≠ Thought:
Language is just a tool for communication

The truth is likely somewhere in between…

The language-thought spectrum

Current evidence points toward the middle: language and thought interact in complex ways, but are not identical.

Evidence: The language network

Note

Fedorenko et al. (2024) - Nature: *The language network as a natural kind*

Key findings:

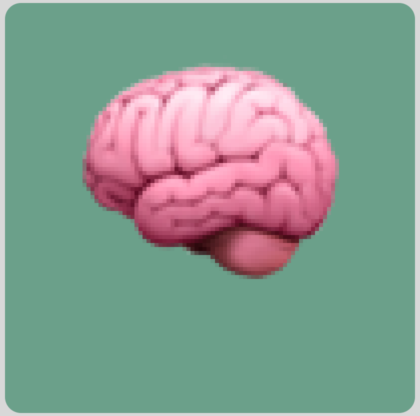
- The brain has a **specialized language network**
- Distinct from: reasoning, math, social cognition, music

Tip

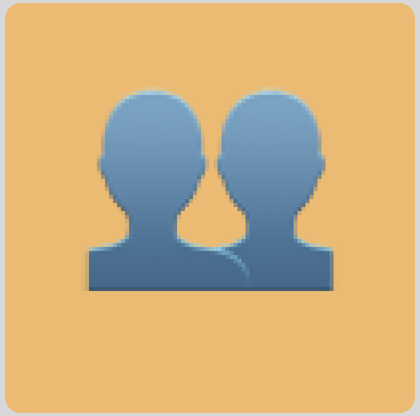
Implication: Language and thought are *separable* in the brain!

Brain networks

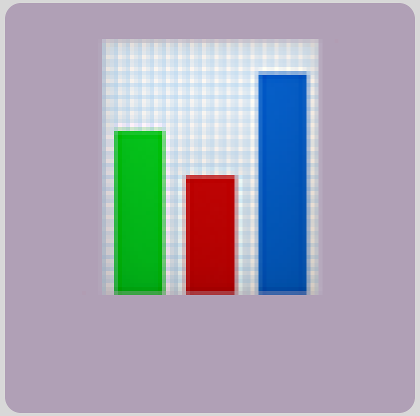
Language



Reasoning



Social



Math

Note

Key insight: Language is a specialized system, not the basis of all thought!

Evidence: Language shapes perception

> **Note**
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> **Lupyan et al. (2020) - TICS:** *Effects of language on visual perception*

Key findings: Having words for things affects how we *see* them

- Speed up visual search
- Alter color perception
- Influence object categorization

The Russian blue example

English speakers:

- One word: "blue"
- Slower to distinguish shades

Russian speakers:

- "siniy" (dark) vs "goluboy" (light)
- Faster to distinguish shades

Tip

Language categories affect *perception*, not just description!

So what about LLMs?

Current evidence suggests:

- LLMs are *very good* at language patterns
- LLMs may have some internal "representations"
- LLMs lack grounding in sensory-motor experience
- No evidence of phenomenal consciousness

> **Tip**
>
> If language and thought are separable, can an LLM have sophisticated language *without*

The grounding problem

Note

Symbol grounding: How do symbols (words) get their meaning?

Humans: Grounded in experience

- See, touch, taste objects
- Act in the world

LLMs: Statistical patterns

- Only see text
- No sensory experience

Our approach

Note

Philosophy of this course: We will build language models *from scratch* to understand what they can and cannot do.

Critical perspective:

- Question assumptions about "understanding"
- Appreciate both capabilities and limitations
- Think about implications for cognitive science

From simple to complex

ELIZA → Classifiers → Embeddings → Attention → GPT

At what point does pattern matching become understanding?

Note

Discussion: At what point (if any) does pattern matching become understanding?

HuggingFace: Your learning companion

We will use HuggingFace for:

- **Course materials:** Free NLP course at huggingface.co/course
- **Pre-trained models:** Download and use state-of-the-art models
- **Datasets:** Access curated datasets for training

> Warning
>
> **Assignment:** Create a free HuggingFace account and complete Chapter 1 of the NLP course!

Next lectures this week

Note

Lecture 2: Pattern matching and ELIZA

- String manipulation
- Regular expressions
- Introduction to ELIZA

Note

Lecture 3 (X-hour): ELIZA implementation

- Complete architecture
- The ELIZA effect
- Assignment 1 details

Readings for this week

Note

Required readings:

1. Weizenbaum (1966): ELIZA
2. Fedorenko et al. (2024): The language network
3. Lupyan et al. (2020): Effects of language on visual perception

Tip: Start with Weizenbaum--it will help you understand the fundamentals!

Key takeaways

1. **Consciousness is complex:** Multiple types, hard to define
2. **Language $\neq$ Thought:** But they interact in interesting ways
3. **LLMs are not conscious:** They are sophisticated pattern matchers
4. **Grounding matters:** Meaning comes from experience
5. **We will build to understand:** Hands-on reveals true capabilities

Questions?

Email

Discord

Moore 349

Note

Office hours: By appointment | **Lab website:** context-lab.com